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Earning Alpha by Avoiding the Index Rebalancing Crowd

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Traditional capitalization-weighted indices generally add stocks with high valuation multiples after persistent outperformance and sell stocks at low valuation multiples after persistent underperformance. It is well known that the price impact of these changes can be large once a change is announced. The subsequent reversal is less well known. For example, in the year after a change in the S&P 500 Index, discretionary deletions beat additions by 22%, on average. Simple rules, such as trading ahead of index funds or delaying reconstitution trades by 3 to 12 months, can add up to 23 basis points a year. This benefit roughly doubles when we cap-weight a portfolio selected based on the fundamental size of a company's business or on its multi-year average market-cap.

Keywords: indexation; optimal trading; passive investing; portfolio construction; transaction costs

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Many arguments in favor of traditional passive index funds are compelling. These funds ostensibly offer low fees, nearly limitless liquidity, very low trading costs, and broad market participation. They closely match the performance of the published index with near-zero tracking error, generally assumed to mean that they incur negligible trading costs. These funds also outperform most active managers most of the time. Although index funds do have low fees and beat most active managers, which are undeniably great advantages, the assertions of nearly limitless liquidity and negligible trading costs are not accurate. In this article, we show that the hidden trading costs absorbed by index investors are shockingly large.[1] We explore alternative index rules to help investors mitigate these costs.

Index management fees and transaction costs have been amply explored in the literature.[2] We demonstrate a source of return drag largely ignored in prior research: that index rebalancing is associated with an additional and hidden cost arising from security selection. Whenever an index rebalances or makes changes to its constituents, the added stocks are routinely priced at a substantial premium to market valuation multiples (i.e., the index buys high), whereas discretionary deletions (with the exception of removals related to mergers, acquisitions, and other corporate actions) are routinely deep-discount value stocks (i.e., the index sells low). In fact, additions tend to be

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priced at valuation multiples more than four times as expensive as those of discretionary deletions, using a blend of relative price-to-earnings (P/E), price-to-cash flow (P/CF), price-to-book (P/B), price-to-sales (P/S), and (if available) price-to-dividends (P/D) ratios.[3] In other words, conventional cap-weighted index funds generally buy high and sell low.

This buy-high/sell-low phenomenon helps explain why from October 1989 through June 2021 the performance of the S&P 500 Index's discretionary deletions beat the additions by an average of over 2,200 bps in the 12 months following an index reconstitution; the difference is statistically significant with a *t*-statistic of 3.40.[4] We begin our analysis in October 1989, the month that the S&P Index Committee began to preannounce changes to the index. This same pattern is evident, but much weaker, in the previous decades.

Index funds inevitably incur some costs that must be borne by the index investor, but investors can be spared some hidden but *avoidable* costs.[5] Unfortunately, index managers have little upside in lowering costs their customers cannot see. Any index fund that seeks to recoup these avoidable costs will necessarily introduce tracking error, a mismatch between the performance of the fund and the performance of the published index. Performance mismatch with the published index is often viewed negatively by investors because most investors have erroneously been led to believe that tracking error relative to the index—*including positive tracking error*—is a sign of sloppiness. This combination of forces creates incentives for managers to *not* try to lower the avoidable index costs, or add incremental return.[6] How can investors solve this problem? In this article, we explore alternative capitalization-weighted index construction rules to help investors mitigate avoidable costs.

One easy tactic is to trade immediately after an index change is announced (the "announcement date"), rather than waiting until the change goes into effect ("effective date"). Another is to delay rebalancing by a year. A third possibility is to choose different stocks to include in the index, selected in a fashion that does not chase fads or abandon deep-value stocks at or near their nadir. An investor can accomplish this in many ways. Two examples are to use stale capitalization weights—tacitly requiring newly large-cap stocks to demonstrate their staying power—or to use fundamental measures of the size of a business to select candidates for the index, then cap-weight the resulting list. After security selection, an investor would then cap-weight the selected stocks to construct a final portfolio very similar to the existing cap-weighted indices, while mitigating the avoidable performance drag from mean reversion.

As direct indexing becomes more common, many investors may opt out of indices that suffer from the return drag induced by buying highfliers and selling unloved value stocks at each rebalancing. Instead, with a few thousand dollars, they can use fractional shares to create their own index fund, complete with the tilts and exclusions they choose and with tax-loss harvesting or any other fine-tuning adjustments they may desire.

Throughout the paper, we use the S&P 500 to illustrate the stock selection problem around rebalancing. We focus on the S&P 500 because it is the most widely followed U.S. equity index. Importantly, most of the large-cap equity indices share the issues we identify with the S&P 500 and consequently share similar opportunities for investors to improve their investment results. We propose a number of methods to enhance index fund returns.

Buy High and Sell Low

Much of the existing literature that examines the costs of indexation focuses on the liquidity effects of index rebalancing. But as we will show, the most important Achilles' heel for index funds is the very avoidable buy-high/sell-low dynamic of adding recent highfliers and dropping deeply out-of-favor stocks, sometimes near a nadir in price; this effect is considerably larger than the liquidity effect. This momentum-chasing selection bias causes investors to lose, on average, tens of basis points of performance annually. While a margin of tens of basis points may seem modest, in the indexing world it is vast. The strategies we explore add anywhere from 12 bps to 46 bps a year to performance. Consider that the average historical one-way turnover of the S&P 500 is 4.4%.[7] Smarter trading strategies that mitigate the indexing world's self-inflicted buy-high/sell-low travails can add roughly 300 to 1,000 bps of performance *a year* when measured relative to this very low rate of turnover. If fund managers will not seek to recapture this avoidable slippage, then index fund investors may seek more direct ways to do so.

Turnover of 4.4% on $7.1 trillion of assets invested in index funds as of 31 December 2021 implies that in any given year, roughly $625 billion in stock transactions takes place, on the bid and ask sides combined, on the days the index rebalances. Much of the trading is concentrated in a very short list of additions and deletions.[8] Roughly $8.5 trillion in additional assets are being benchmarked to the S&P 500, and many of the benchmark-hugging active managers—as well as hedge funds that front run the preannounced index fund trades—may also jump on the same trades as the index funds.[9] Honghui, Noronha, and Singal (2006) estimate that around the time of their article publication, investors in S&P 500 and Russell 2000 indices alone were losing between $1.0 billion and $2.1 billion a year from the two indices combined. Given the subsequent significant growth in passive assets these costs have likely gone up.

In order to evaluate the costs associated with index implementation, it is important to understand the practical details associated with index rebalancing. While originally S&P Index changes were made public on the effective day of index change, in October 1989, S&P began preannouncing changes to the index along with the effective date for when those changes will be reflected in the index, which could be days or weeks after the announcement date. Until February 2017, changes in index holdings were made at the market's closing price on the effective date. Since then, changes have been made at the prior close (effective date minus 1). We will refer to the date on which the closing price is the basis for the change in the S&P 500 as the "trade date." Under either convention, the trade date is the peak volume day, when most index fund managers trade at or near the market close to avoid tracking error. Trading volume is often also very large on the previous and subsequent day.[10]

The trading costs of index funds are masked because they are already borne by the published index itself. During the grace period, the price impact—no matter how large—will affect equally the performance of both the index fund and the index against which it is measured. Thus, index funds need not suffer underperformance relative to the index from the price impact of their own trades. This outcome is better interpreted as near-zero tracking error *against an index that has been altered to permit near-zero tracking error*. Unfortunately, near-zero tracking error is often falsely interpreted as evidence of near-zero trading costs.

The dirty secret of index fund investing is that the transaction costs are still there and simply hidden in plain sight.[11] The closer a manager trades to the closing price on the trade date, the closer the fund will track the index. Because an index fund's mandate is to track the index, and because these funds largely compete on fees, most index fund managers today are far more interested in reducing tracking error than in adding value.[12]

As we demonstrate later in more detail, if S&P did not preannounce index changes, the hypothetical index performance would have been 20 bps a year better. No index fund could match this performance without trading at the market close *preceding* the announced index change. Accordingly, most index funds would likely have lagged the index by roughly that margin (with tracking error of about 31 bps a year).[13] In the pre-1989 period, when Standard & Poor's announced changes in the S&P 500 after the market had closed, with those changes taking effect at the already-known closing price on that day, significant trading-related impact costs would show up as slippage against the theoretical index. After 1989, with changes preannounced, any trade impact costs are baked into index performance, so that the index fund trading costs, and the associated tracking error, seem to disappear. Of course, investors still experience tracking error and slippage from the theoretical (and no longer published) index, following the old rule of announcing an index change after the close of trading.

As a first step in estimating the true cost of indexing, we constructed a history of the S&P 500's historical constituent changes.[14] The sample of historical component changes consists of 1,196 additions and 1,194 deletions from March 1970 to June 2021.[15] Further, we use stock return data from CRSP as a proxy to determine the nature of the changes: nondiscretionary changes due to mergers, spin-offs, or acquisitions and discretionary changes made by the S&P Index Committee following the S&P 500 guidelines.[16]

Our analysis focuses primarily on the period after 1 October 1989. Over this span, roughly 90% of additions are discretionary, with just 116 being nondiscretionary (e.g., company spin-offs and divestitures, such as 3Com spinning off Palm in 2000). In contrast, more than 60% of the index deletions are nondiscretionary (e.g., bankruptcies and mergers).[17]

The much larger number of nondiscretionary deletions compared to nondiscretionary additions reflects corporate actions: Large companies are more likely to consolidate their businesses or to be acquired (the main sources of nondiscretionary deletions) compared to spinning off a large segment of their business (the main source of nondiscretionary additions). We provide more detail on the nature of additions and deletions by subperiod in Table 1.

When we examine preannouncement stock performance, as reported in Table 2, Panel A, we observe

Table 1. S&P 500 Historical Constituent Changes, March 1970–June 2021

	Additions			Deletions		
Date range	Discretionary	Nondiscretionary	Total	Discretionary	Nondiscretionary	Total
Mar 1970–Jun 2021	1,080	116	1,196	464	730	1,194
Mar 1970–Sep 1989	417	26	443	165	278	443
Oct 1989–Jun 2021	663	90	753	299	452	751

Source: Research Affiliates, LLC, using data from Siblis Research and CRSP, cross-referenced against Wikipedia. We use constituent change data from Siblis for the period October 1989–March 2017 and data from Wikipedia for the period April 2017–June 2021.

that, on average, additions outperform deletions with the performance gap between additions and deletions monotonically accumulating to an average difference of 70.6% over the 12 months prior to the announcement. New additions outperform the market by an average of 41.5% prior to the announcement they are being added to the index; meanwhile, discretionary deletions underperform the S&P 500 by −29.1%. This magnitude of over- and underperformance presumably helps to draw the attention of the S&P Index Committee. The difference in performance between additions and deletions is highly statistically significant with a *t*-statistic of 13.98.

From the announcement date to the market close on the trade date, as shown in Table 2, Panel B, additions on average appreciate by another 5.0% relative to the market, while deletions trail the market by an additional −7.2%. The result is a performance gap over the period of 12.2%. The price movement around the announcement is highly statistically significant with a *t*-statistic of 14.60.[18] About one-fourth of this gap accrues on the trade date itself when the majority of the index-fund trading occurs.[19]

Notably, this price movement immediately precedes the announcement date and continues until one day after the trade date (averaged across hundreds of additions and discretionary deletions). While we might surmise that the selection of discretionary additions and deletions is partly motivated by the performance spread between the additions (past winners) and deletions (past losers), this hypothesis would not explain the impressive performance gap in the week—and even on the final day—before the *announcement* is made. Panel A shows that the one-year performance gap is an already-substantial 21 bps a day (accumulating to 70.6% a year); the six-month and three-month gaps are of similar magnitude. This relative performance difference is likely a major factor in the S&P Index Committee's decision to add or drop a stock. The gap widens fivefold to 5.5% in the week before the announcement and tenfold to 2.4% on the day before the announcement.[20]

On the day after the trade date, as reported in Table 2, Panel C, we find another 1.2% performance spread in favor of the additions over the deletions, perhaps due to catch-up trades by index funds that did not complete all of their addition and deletion trades on or before the trade date. If we add the 2.4% performance spread on the day before the announcement *and* the 1.2% spread the day after the trade date, the 12.2% performance spread between additions and deletions soars to 15.8%.

In the first year following the trade date (we use 252 equity-market trading days), additions suffered a modest performance drag of −1.6%, while the deletions outperformed the market by 20.4%, a performance difference of 22.0% (Panel C). The difference is statistically significant with a *t*-statistic of 3.40. If we exclude the first trading day after the trade date, when additions continue to outpace deletions, discretionary deletions outperform additions by more than 23%.[21]

The magnitude of our findings is largely in line with the literature. In the period before S&P began to preannounce index changes, Shleifer (1986) found a 2.8% abnormal return for additions to the index (from September 1976 to December 1983). Arnott and Vincent (1986) examined a longer period and found a 2.7% return for additions and −11.5% for deletions over the 20 days after a stock was added to the index. Lynch and Mendenhall (1997) showed that, after Standard & Poor's started preannouncing index changes, much of the effect shifted to the grace period, peaking on the announcement date. For the period from announcement to about a week after the change, they demonstrated an abnormal average return of 4.9% for additions and −15.7% for deletions. Elliott et al. (2006) examined the announcement effect for the index additions, which they

Table 2. Relative Performance Over Various Horizons, Average Cumulative Return Relative to Market, October 1989–June 2021

A. Relative Performance before Announcement

Days before announcement	Additions	Discretionary deletions	Nondiscretionary deletions	Additions minus discretionary deletions	Adds Minus Deletions *t*-Statistic
252 (1 year)	41.48%	−29.11%	21.29%	**70.59%**	**13.98****
126 (6 months)	16.35%	−18.81%	14.96%	**35.16%**	**12.98****
63 (1 quarter)	6.13%	−12.79%	5.74%	**18.91%**	**10.52****
21 (1 month)	2.13%	−6.42%	1.21%	**8.55%**	**7.41****
5 (1 week)	0.95%	−4.51%	−0.89%	**5.46%**	**6.55****
1 Day	0.16%	−2.20%	−0.49%	**2.37%**	**3.87****

B. Relative Performance between Announcement and Trade Date

Cumulative between announcement and trade date	Additions	Discretionary deletions	Nondiscretionary deletions	Additions minus discretionary deletions	Adds Minus Deletions *t*-Statistic
Close of trade date − 1 day	3.84%	−5.22%	−0.15%	**9.06%**	**12.92****
Close of trade date	4.99%	−7.23%	−0.65%	**12.21%**	**14.60****

C. Relative performance after trade date

Days after trade date	Additions	Discretionary deletions	Nondiscretionary deletions	Additions minus discretionary deletions	Adds Minus Deletions *t*-Statistic
1 day	0.67%	−0.49%	–	**1.16%**	**2.83****
5 (1 week)	0.06%	0.86%	–	**−0.81%**	**−0.69**
21 (1 month)	−0.27%	7.99%	–	**−8.26%**	**−3.23****
63 (1 quarter)	0.55%	13.26%	–	**−12.71%**	**−2.97****
126 (6 months)	0.44%	15.05%	–	**−14.61%**	**−2.45***
252 (1 year)	−1.61%	20.43%	–	**−22.03%**	**−3.40****

Note: We calculate the *t*-statistics for the additions minus deletions by pairing the same-day highest ranking by change-day market-cap, ranking additions with deletions and take the difference between the two. If there is no corresponding pair, we take the difference with the S&P 500 return.
Source: Research Affiliates, LLC, based on data from Siblis Research, Wikipedia, and CRSP.

found to be highly statistically significant (*t*-statistic of 21.88) and attributed the effect largely to increased investor awareness. Sui (2006) found an 8.4% return for additions and −11.1% for deletions for the period from the announcement date to the effective date. Sui went one step further to determine short-term mean reversion of −2.3% for additions and 4.9% for deletions over the 20 days after the effective date.[22] We are not aware of any prior work that focused on the year (or longer) before and after index reconstitution.[23]

The post-rebalancing reversal of performance for the additions and deletions is unsurprising when we examine the valuation ratios of additions and deletions relative to the market, as shown in Table 3.[24] The additions, based on an average of P/B, P/E, P/CF, P/S, and P/D, are 92% more expensive than the market. The discretionary deletions, in contrast, are 55% cheaper than the market based on the combination of the five relative-valuation measures. The additions are therefore 4.3 times as expensive as the discretionary deletions, on average. With a roughly fourfold relative valuation ratio, the performance spread of more than 20% between the additions and discretionary deletions in the subsequent year is an unsurprising combination of the value effect and mean reversion.

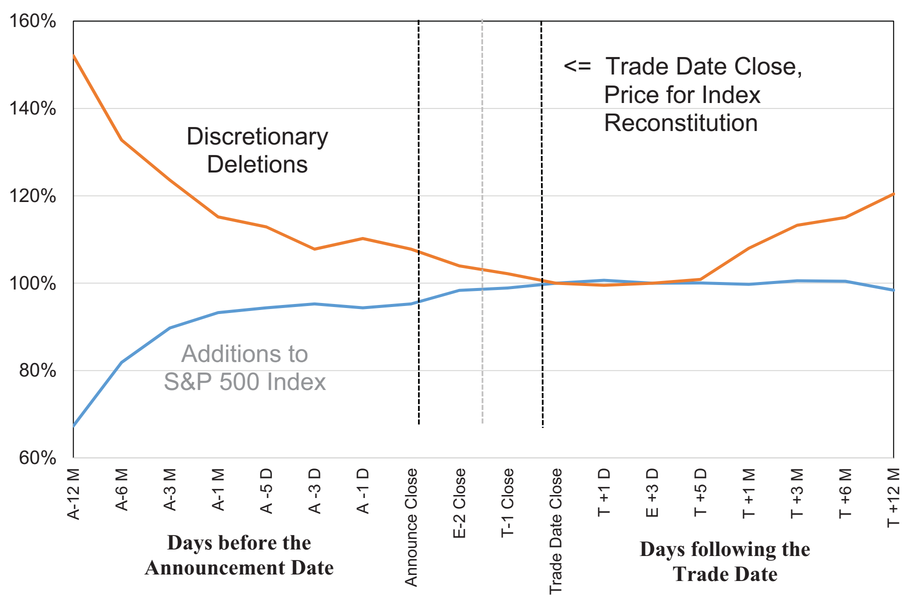
Figure 1 graphically represents the results in Table 2, illustrating the buy-high/sell-low pattern of the S&P 500 for the period October 1989 through June 2021. The stocks added to the index outperform the

Table 3. S&P 500 Additions and Deletions, Average Valuation Relative to the Market, October 1989–June 2021

Type	P/B	P/E	P/CF	P/S	P/D	Average	Count
Additions	1.55	1.83	2.11	2.21	1.91	**1.92**	753
Deletions	0.58	0.74	0.62	0.52	0.67	**0.62**	751
Discretionary deletions	0.42	0.53	0.45	0.34	0.50	**0.45**	299
Nondiscretionary deletions	0.92	1.09	1.01	0.99	1.00	**1.00**	452
Additions relative to discretionary deletions	3.72	3.45	4.72	6.54	3.80	4.30	

Source: Research Affiliates, LLC, based on data from Siblis Research and CRSP, cross-checked against Wikipedia.
P/B = price-to-book ratio; P/E = price-to-earnings ratio; P/CF = price-to-cash flow ratio; P/S = price-to-sales ratio; P/D = price-to-dividends ratio.

Figure 1. S&P 500 Additions and Discretionary Deletions' Performance Relative to Market, October 1989–June 2021



Source: Research Affiliates, LLC, based on data from Siblis Research, Wikipedia, and CRSP.

market, on average, by 41.5% in the period covering 12 months before the announcement until the index adds these names, whereas the stocks sold by the index lag the market by −29.1% over the same period. In the year after the trade date, the situation reverses and the deletions outperform the additions by over 22%, with the lion's share of the difference coming from the deletions.

When we break the 1989–2021 sample into its two halves, we find the mean-reversion effect gets stronger in the second half of the sample for the additions: −1.2% in the first half of the sample and −2.5% in the second. The opposite is true for the deletions, however, falling from 31.8% in the first half to 12.3% in the second. The one-year spread between additions and deletions is, therefore, 33.0% from 1989 to 2004 and 14.8% from 2005 to 2021.[25] Both figures are statistically significant, but the difference between them is not. Accordingly, the diminished spread between additions and deletions is consistent with the findings of Soe and Dash (2010) and Preston and Soe (2021), who found declining index effects in the latter part of the sample. Perhaps the market is becoming more efficient around index rebalancing or perhaps the absence of a value effect during most of the 2005–2021 span explains the diminishing spread.

We can attribute the post-rebalancing returns for additions and deletions to three effects: announcement effect, liquidity effect, and mean-reversion effect. If we start a day before the announcement (to allow for potential information leakage or research-based anticipation by liquidity providers) and track additions for the 12 months following the trade date, the additions outperform the index by 3.5%.[26] Mean reversion after the trade date is less than half this amount and therefore is dominated by the announcement and liquidity effects. Deletions outperform by 11.0%[27] from the day before the deletion is announced until a year after the trade date. The mean reversion after the trade date is twice as large as the total, so for deletions, the mean-reversion effect dominates the other two effects.

An Improved Market Index

Using the S&P 500, we have shown that the buy-high/sell-low dynamic of traditional large-cap indices hurts investors in two ways: the price impact of many billions of dollars in stocks being traded on index trade dates as well as a subsequent pattern of mean reversion in the affected stock's relative performance. Broader indices, such as the Russell 1000 and Russell 3000, exhibit the same effect. Companies that are added, and deletions that are not a consequence of corporate actions, tend to be smaller with these broader indices, so even like-size moves before and after the trade date will not have as much portfolio performance impact. While the effect is smaller, it is as persistent and as statistically significant.

How large are the impacts of these two forces on index returns and investor wealth? To answer this question, we simulated four indices based on the S&P 500 and progressively introduced changes to each of the simulated indices to gauge the effect of each change. The four simulated indices are (1) a replication of the S&P 500; (2) a replication that makes the index changes on the announcement date, as was the case for the S&P 500 prior to October 1989; (3) another replication that makes the index changes on the day after the announcement; and (4) a replication that delays trades by a 3- or a 12-month lag—which we call "lazy rebalancing"—after a large portion of the mean reversion in price has taken place.

Actual S&P 500. Stocks in the index are weighted based on their market capitalization, adjusted for float to reflect the portion of the market capitalization available to the general public. The float adjustment was introduced in March 2005 and fully transitioned in September 2005, prompted by the growing number of new tech companies for which the founders retained a large closely held stake. Before the change, the index weights were based on simple market capitalization, adjusted from time to time to adjust for stock buybacks and secondary equity offerings.

Replicated S&P 500. We do not have access to the exact historical weights for the individual stocks in the S&P 500 or to their exact float adjustments. Therefore, we constructed a replicated version of the index, in order to compare strategies on an apples-to-apples basis.[28] We began our replication of the S&P 500 by working backward from a snapshot of SPY ETF holdings on 30 June 2021. Then, using the history of constituent changes as a guide, we roll backward in time over the past 32 years to October 1989, periodically cross-checking the simulated holdings against the holdings of the SPY ETF (or the Vanguard 500, for the years before the SPY ETF was launched).[29] In our replication periods, as we roll back before September 2005, we shift from a weight based on the float-adjusted S&P 500 to a weight based on market capitalization. The replicated S&P 500 matches the actual S&P 500 with an R^2 of 0.9995.[30]

Replicated Trade-on-Announcement S&P 500. Before October 1989, changes to the S&P 500 were executed at the close of the announcement day. We wanted to know how the S&P 500 would have performed since that date if the S&P Index Committee still followed this protocol. No index fund manager could do this because they would need to know the index change before it happens, but the index calculator certainly can do this, exactly as they did for over 30 years before October 1989. To gauge the performance impact of the grace period, we computed the replicated trade-on-announcement S&P 500 by modifying the replicated S&P 500 for rebalancing to occur at the close of the announcement date instead of at the close of the trade date.

While this strategy cannot be replicated with actual money, the exercise is still valuable for analytical purposes. This replication basically shows the performance for the S&P 500 *as if the S&P Index Committee had not begun in 1989 to preannounce index changes*. This exercise provides a useful, if crude, approximation of the real-world trading costs that index fund investors incur to execute index constituent changes.

Replicated Day-after-Announcement S&P 500. While a trade-on-announcement strategy cannot be replicated with actual money, a trade-day-after-announcement strategy shows the performance an index fund manager could have earned by trading quickly. Of course, if all indexers did this, the day-after-announcement price would have been changed by these large block trades, perhaps moving share prices all the way to the closing price on the trade date. Notably, however, a small, nimble index fund manager who is willing to accept 29 bps of tracking error relative to the index could have captured this 12 bps advantage.

Replicated S&P 500 with Delayed Rebalancing. The actual S&P 500 adds new stocks at high valuation multiples and drops them at deeply depressed valuations. If the share prices and valuation multiples subsequently revert toward the market mean, as they empirically tend to do, index performance suffers. We simulated a replicated S&P 500 that delays trades by several months (we show results for 3 and 12 months), after a large portion of the mean reversion in price has occurred. Delaying trading inevitably causes tracking error versus the actual S&P 500. Unlike the trade-on-announcement replication, this lazy replication index is easy to implement. The difference in its performance versus the replicated S&P 500 captures the impact of mean reversion on index performance.[31]

Lazy Is Good for Index Fund Management! We display the performance characteristics of the replicated indices from October 1989 through June 2021 in Table 4 and their respective dollar-growth paths relative to the replicated S&P 500 in Figure 2.

We use the replicated index rather than the actual S&P 500 as the benchmark to examine the performance implications of leading or lagging the rebalancing decisions. Given that we do not have the precise portfolio weights of the actual index, using the replicated index makes our analysis an apples-to-apples comparison while minimizing the impact of any other hard-to-replicate features of the index. For other comparisons, in which we introduce alternative lists of 500 stocks, we use the actual S&P 500.

Figure 2 graphically illustrates the efficacy of these simple approaches to improve index fund results, the consistency of results, and the times when lazy strategies fail (notably, during growth-dominated bubbles, despite the persistent positive four-factor alpha in most periods).[32]

We make several observations based on our results:

- The replicated trade-on-announcement S&P 500 outperforms the replicated S&P 500 by 20 bps a year (10.68% versus 10.48%). Of course, the full 20 bps are hard to capture, because doing so would require advance knowledge of Standard and Poor's index reconstitution decisions before they are made public. But fully 60% of that spread—12 bps a year—can be captured by rebalancing the index on the day after the announcement. The 20 bps difference is a reasonable gauge of the cumulative price impact from index fund trading. Although index funds may trade primarily at the close on the trade date, hedge funds and market makers will have imposed the index funds' trading costs on the published index before most index fund managers trade on the trade date.
- The lazy replicated S&P 500, delayed by 3 months and 12 months, outperforms the replicated S&P 500 by 13 bps and 23 bps, respectively. The difference in the returns reflects part of the return drag experienced by S&P 500 investors due to the buy-high/sell-low dynamic.
- Each of these reduces the volatility of the S&P 500 by a small but statistically significant margin.

The benefit from lazy indexing, for which we delay the trade by a year, is nearly identical to the benefit of clairvoyant trading, in which we trade just before Standard & Poor's announces a change in the index. We should point out that earning clairvoyant alpha is not as implausible as it might seem. Let us not forget the substantial movement in the single day before an announced change, which suggests that some people—presumably proactive indexers and hedge funds—are doing their homework and are able to anticipate many of the changes *with the result that much of the 20 bps can presumably be captured.*[33]

We note that while index fund managers appear to care deeply about every basis point of tracking error relative to the published index, that index has a 31 bps annual tracking error relative to its own trade-on-announcement S&P 500, which is the way the index was calculated before October 1989. Most of the early- and late-trading variants have no more tracking error than the S&P 500 has relative to its own trade-on-announcement variant.

Figure 2 shows how much an S&P Index fund investor can gain by trading early or late, because of the fund's fixation on zero tracking error. These

Table 4. Performance of Various Replicated S&P 500 Capitalization-Weighted Indices, October 1989–June 2021

A. Risk and Return Characteristics

Index	Return	Volatility	Sharpe Ratio	Excess Return vs. Replicated S&P 500	Tracking Error vs. Replicated S&P 500	Information Ratio vs. Replicated S&P 500	*t*-Statistic on Excess Return
Replicated S&P	10.48%	14.42%	0.54	0.00%	0.00%	0.00	—
Replicated Trade-on-Announcement S&P	10.68%	14.36%	0.56	0.20%	0.31%	0.64	3.63**
Replicated Day-after-Announcement S&P	10.61%	14.37%	0.55	0.12%	0.29%	0.42	2.37*
Lazy Replicated S&P, Trades Delayed by 3 Months	10.61%	14.34%	0.56	0.13%	0.33%	0.39	2.17*
Lazy Replicated S&P, Trades Delayed by 12 Months	10.71%	14.29%	0.56	0.23%	0.51%	0.46	2.57*

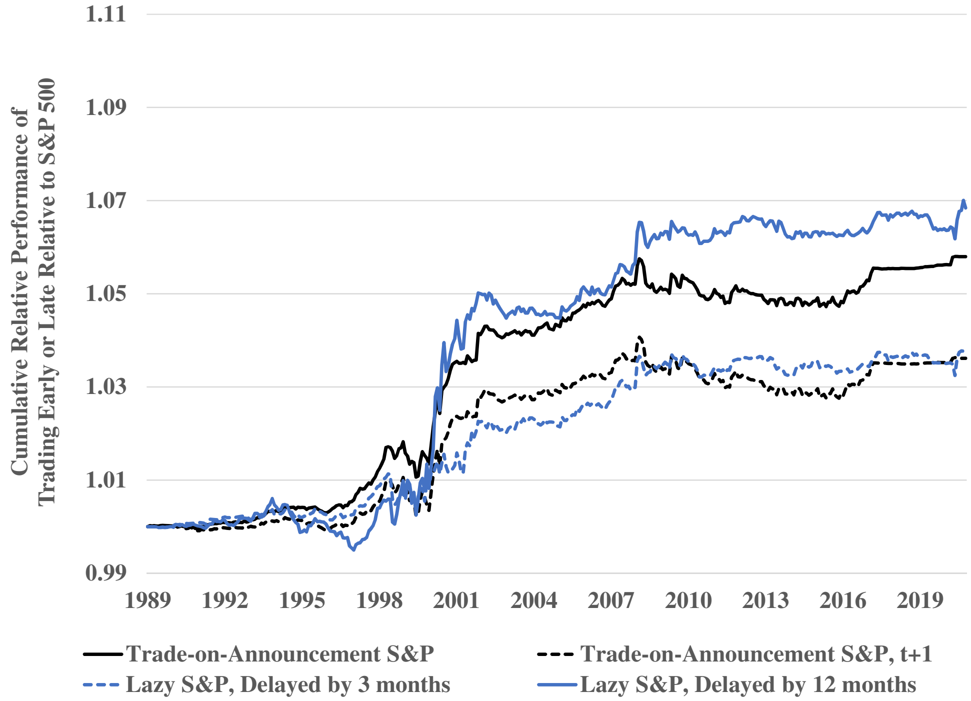
B. CAPM and Four-Factor Model Based Performance Characteristics

		CAPM Model	Fama-French Three-Factor Model plus Momentum				
				Coefficients, Factor Loadings			
Index	Excess Return vs. S&P 500	Annual Alpha	Annual Alpha	Market[a]	Size	Value	Momentum
Replicated Trade on Announcement S&P	0.20%	0.21%	0.19%	0.997	−0.005	0.007	0.001
t-statistic	*3.63***	*3.85***	*3.69***	*−2.98***	*−3.64***	*4.98***	*1.09*
Replicated Day-after-Announcement S&P	0.12%	0.14%	0.13%	0.997	−0.005	0.007	0.001
t-statistic	*2.37**	*2.69***	*2.54**	*−3.04***	*−3.87***	*5.09***	*0.82*
Lazy Replicated S&P Delayed by 3 month	0.13%	0.15%	0.14%	0.995	−0.005	0.009	0.001
t-statistic	*2.17**	*2.65***	*2.48**	*−4.02***	*−3.41***	*6.15***	*0.88*
Lazy Replicated S&P Delayed by 12 month	0.23%	0.27%	0.23%	0.992	−0.008	0.022	0.002
t-statistic	*2.57**	*3.15***	*3.10***	*−4.9***	*−4.04***	*11.01***	*1.39*

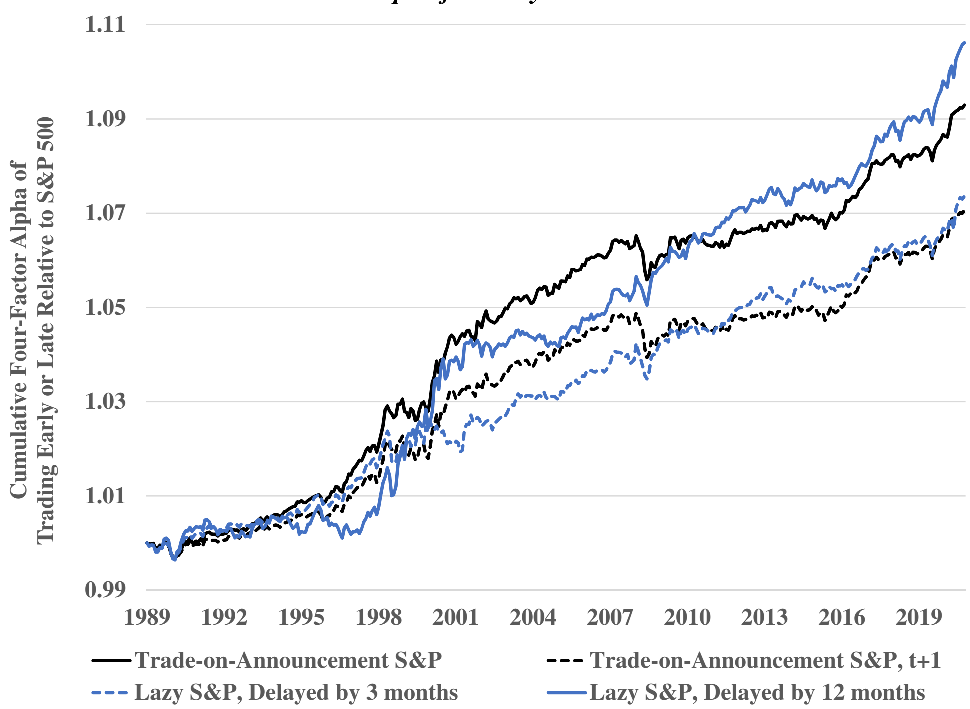
[a]For the market beta, the *t*-statistic measures significance of the difference from a neutral beta of 1.00.
** and * indicate significance at the 1% and 5% level, respectively. Please refer to text for descriptions of the strategies. We use the replicated S&P 500 return to proxy the market factor.
Source: Research Affiliates, LLC, based on data from Siblis Research, Wikipedia, and CRSP.

Figure 2. Cumulative Excess Return of Indices that Use a Different Date for Index Reconstitution Relative to S&P 500, October 1989–June 2021

A. Early or Lazy Index Reconstitution



B. Cumulative Four-Factor Alpha for Early or Late Index Reconstitution



Note: Please refer to the text for descriptions of the strategies. Panel B four-factor returns use the Fama and French (1993) three-factor plus momentum model.
Source: Research Affiliates, LLC, based on data from Siblis Research, Wikipedia, and CRSP.

alternative strategies, each of which makes identically the same trades as the S&P 500, but does so early (on the announcement date or one day later) or lazy and late (3 and 12 months after the trade date), add 12 to 23 bps a year relative to the replicated S&P 500. The cumulative effect leaves our index fund investors 3.6% to 6.8% richer, as long as they are willing to accept some tracking error—and *less* overall beta and risk—relative to the published index.

This potential source of index fund alpha is even larger than it seems once we adjust for portfolio factor loadings. Stocks that are added a bit early or late are slightly less expensive highfliers, while discretionary deletions are slightly less deeply discounted. This means that, in a growth dominated market, these early- and late-trading strategies would do well to merely match the replicated S&P 500. The comparison shown in Table 4, Panel B, and in Figure 2, Panel B, is easy to do by regressing each return series, relative to the replicated S&P 500, against a four-factor model (the 1993 Fama-French beta, size, and value factors plus momentum).

Because the portfolio is identically the same as the replicated S&P 500, albeit with trading shifted a little early or late, these style difference are very small, yet they are large enough to matter. These early- or late-trading strategies all have a trivial momentum tilt, and a modest tilt toward lower beta, larger size, and stronger value. While small, the value tilt is the strongest factor loading, with the highest statistical significance, in every case. If a herd of elephants is trying to go through a revolving door all at the same time, we benefit by choosing to enter before or after they do their damage.

Adjusted for the style differences, the multi-factor alpha is larger than the simple return difference. After 32 years, the investor in these strategies is wealthier than the investor in the replicated S&P 500 by 7.1% to 10.6%, on a four-factor risk-adjusted basis. Also, the style tilts explain why these strategies show relatively flat performance over the last dozen years when value (for the most part) significantly underperformed growth. Table 4, Panel B, and Figure 2, Panel B, show that the cumulative four-factor alpha is remarkably persistent throughout the full period, with impressive statistical significance.[34]

More Ways to Add Value. According to Morningstar, as of December 2016 the price competition between index funds had been reduced to a single basis point (0.01%) for retail products and to fractions of a basis point for institutional clients. On one level, this makes sense. Being able to shave just a single basis point off the costs an investor incurs is a boost to the investors' bottom line. For instance, on the $7.1 trillion in assets tracking the S&P 500, one basis point fewer in fees saves investors $710 million, *as long as the fee cut makes no difference in performance before fees*. That said, if some managers can avoid hidden costs that are more than an order of magnitude larger, any focus on fee differences as small as a single basis point is rather foolish.

Some vendors even offer retail products with zero fees. These vendors are not charitable organizations, happy to invest time and effort for no recompense. It should seem obvious they must have non-fee sources of revenue, such as securities lending, or cross-marketing into more lucrative products. Because these revenue sources dwarf the fee differences, the folly of obsessing over tiny fee differences becomes even more self-evident.

What if one manager cares more about minimizing tracking error than about generating alpha and another manager cares about both? One saves their investors one basis point in annual fees; the other earns their investors 12 to 23 bps more each year by trading early or late relative to the index. This does not make conventional indexing bad—active management fees dwarf this $9 billion to $16 billion hidden cost—it just means indexing can be materially improved.

We harbor no illusions that this 12 to 23 bps benefit can be earned if all indexers pursue these trade-early or trade-late strategies, because doing so merely moves some of the price impact earlier or later than the time period in which it is currently incurred, from announcement date to the trade date. Nonetheless, much of this potential benefit is presumably being pocketed by hedge funds and market makers who are trading ahead of the index funds. Why should index fund investors not share in this benefit?

Revisiting Which 500 Stocks to Cap-Weight. We have already explored ways to mitigate the return drag from trading before the trade date or from waiting 3 to 12 months to effect the index trades—but there is a better way to alleviate that return drag. Instead of buying stocks that have recently soared in price and selling those that have recently plunged, an investor can choose to cap-weight a different portfolio of 500 stocks, chosen based on measures of company size that are less sensitive to recent price movements. For example, not unlike the strategies explored by Arnott, Beck, and Kalesnik (2015), an investor can use a multi-year average of market capitalization as a relevant measure of company size.[35] An added benefit of using the average of multi-year market capitalization is lower turnover.

Better still, if we select the 500 stocks based on the fundamental size of a business—using its footprint in the macroeconomy rather than its market value—and then cap-weight the list, results are even more powerful.

To demonstrate how simple index construction techniques can reduce the negative consequences associated with the buy-high/sell-low dynamic and the turnover costs of cap-weighted indices, we simulated the following six index strategies and compared them to the S&P 500.

Top 500 by Market-Cap. The S&P 500 aims to hold 500 of the largest market-cap stocks, but the S&P Index Committee does not rigidly adhere to this goal. Since 1989, at any given time, the S&P 500 has excluded on average 120 stocks that rank among the 500 largest market-cap stocks and included 120 smaller-cap stocks that do not rank in the top 500. Here, we consider an index strategy that is actually composed of the 500 largest U.S. stocks by market capitalization, rebalanced annually, with constituent stocks weighted based on market capitalization.

Top 500 by Five-Year-Average Market-Cap. An alternative 500-stock index strategy consists of the 500 largest U.S. stocks selected by *five-year-average* market capitalization but weighted based on current market capitalization and rebalanced annually. The result is that a stock whose price and market capitalization have risen sufficiently to qualify it for the previous top 500 index must "season" for a while and maintain the qualifying market-cap—on average for five years—before being added to the index. The reciprocal holds true for stocks that have briefly tumbled out of the top 500 list.[36]

Top 500 by Fundamental Size. The index selects the 500 largest public companies, chosen based on the fundamental economic scale of the business, then weights the companies based on current market capitalization similar to other cap-weighted indices. Like the top 500 by market-cap, the portfolio is rebalanced annually. For the purposes of this study, a company's fundamental economic size is measured as a blend of company book value adjusted for intangibles, past-five-year-average sales adjusted for company debt-to-asset ratio, past-five-year-average cash flow adjusted for company R&D expenses, and past-five-year-average of dividends plus share repurchases, each measured as a percentage of the publicly traded universe. Astute readers will note that this construction technique is quite simply a 500-stock fundamentals-based index, reweighted by market capitalization.[37]

Leaving aside corporate actions and focusing on discretionary additions and deletions, a stock is added to the index because the company's business has grown, not because its share price has soared. And a stock is removed from the index because the company's business has eroded so that it no longer qualifies for the index, not because its share price is in free fall. We use the five-year-average metric to measure the scale of the business. Although changes in a stock's price and changes in the size of the company's business are correlated, the correlation is not strong. The valuations of additions are slightly elevated at about 1.3 times relative to the valuations of discretionary deletions in contrast to the roughly four-fold relative valuation difference observed in the actual S&P 500.

Each of the Three Preceding Methods, with Banding and Seasoning at 20%. *Banding* is a technique that lessens the sensitivity of turnover to small changes in a company's rankings around the target boundary. For example, banding at 20% means that a stock held in the portfolio would need to drop in size to a rank of 600 or higher to be excluded from the portfolio and would need to increase in size to a rank of 400 or lower to guarantee its inclusion in the portfolio. *Seasoning* means that a stock needs to rank at 400 or lower *for two consecutive years* in order to be added to the index or to rank at 600 or higher for two consecutive years in order to be dropped.[38] These stipulations largely eliminate the propensity to buy the latest highflier just before it tanks or to sell an unloved stock in free fall just before it rebounds. They also lower turnover by reducing the risk of reversing the trade a year or two later. The index is rebalanced annually and its constituents are weighted based on market capitalization.

Critics will likely observe that, with all six of these hypothetical 500-stock index fund portfolios, we are changing the composition of the index and therefore creating an entirely different index. *That is the point!* The S&P 500 is not, and has never been, an index of the 500 largest market-cap stocks trading in the U.S. market. It has, on average, about 380 stocks that are among the 500 U.S. stocks with the largest market value and 120 stocks that are *not*, including a handful of companies that are not U.S.-domiciled businesses. By selecting the 500 companies based on the fundamental size of the company's business, the result is nearly identical with about 370 stocks, on average, that are among the 500 most-valuable companies based on market capitalization. If choosing index constituents based on the fundamental size of the business has essentially identical representation among the 500 largest market-cap stocks, when compared with the choices made by the S&P Index Committee, how can a critic persuasively say that one construction technique is correct and the other is not?[39]

Why Alternative 500-Stock Universes Add Value. We compare the performance characteristics of the six preceding index strategies with the S&P 500 in Table 5 and make the following observations based on our results:

- The top 500 by market-cap strategy outperforms the S&P 500 by 25 bps a year with annual turnover of 5.4%, a bit higher than the turnover of the S&P 500. *We find it startling that the S&P 500 has larger tracking error when compared to the simple 500 largest by market-cap strategy than against any of the other five strategies*. This comparison vividly illustrates the fact that the S&P 500 is *not* composed of the 500 largest-cap U.S. stocks as many casual observers believe it to be.
- The top 500 by five-year-average market-cap strategy outperforms the S&P 500 by 38 bps a year (10.92% versus 10.54%) from October 1989 to June 2021, with a turnover nearly identical to the turnover of the S&P 500. It seems that using long-run averages to decide which stocks to add to an index and which to delete is an effective way to remove the buy-high/sell-low dynamic inherent in a cap-weighted index. Importantly, when we compute the weight of the stocks, we do not filter out the stocks that had recently undergone an IPO or lack the market capitalization history for some other reason. For these stocks, we simply use the available history.[40]
- The top 500 by fundamental-size strategy outperforms the S&P 500 by 46 bps, the highest magnitude of outperformance among the variants we tested. This evidence validates the idea

Table 5. Performance of Alternative Index Selection Metrics of Top 500 Stocks, October 1989–June 2021

Index	Return	Volatility	Sharpe Ratio	Value-Add vs. S&P 500	Tracking Error vs. S&P 500	Information Ratio vs. S&P 500	Annual Turnover
S&P 500	10.54%	14.57%	0.54	—	—	—	4.4%
Top 500 by Market-Cap	10.78	14.64	0.56	0.25	1.12	0.22	5.4
Top 500 by Five-Year Avg Market-Cap	10.92	14.42	0.57	0.38	0.75	0.51	4.5
Top 500 by Fundamental Size	11.00	14.25	0.59	0.46	0.99	0.47	4.4
Top 500 by Market-Cap, With Banding & Seasoning at 20%	10.81	14.58	0.56	0.27	0.92	0.29	4.4
Top 500 by Five-Year Avg Market-Cap, With Banding & Seasoning at 20%	10.94	14.32	0.58	0.41	0.76	0.54	4.0
Top 500 by Fundamental Size, With Banding & Seasoning at 20%	10.99	14.23	0.59	0.46	1.01	0.45	4.1

Note: Please refer to the text for descriptions of the strategies.
Source: Research Affiliates, LLC, based on data from Compustat and CRSP.

that using a company's fundamental size to decide an index's composition is a powerful remedy for the buy-high/sell-low drawback of a traditional cap-weighted index, which selects stocks based on market capitalization.

- Banding and seasoning can be quite effective at further lowering turnover. When both banding and seasoning are applied to any of our hypothetical indexing strategies, annual turnover is at or below the S&P 500's turnover in all cases. Banding and seasoning reduce portfolio risk in all cases, albeit only modestly, without impairing performance.[41]

If we are no longer chasing recent performance in selecting companies to add or remove from the index, then our additions will have less of a growth bias and less of a momentum bias relative to the deletions. As highflier additions to the index also often have a high beta, a different selection method will also lead to a slightly lower-beta portfolio. Table 6 provides a deeper dive as we examine performance either in a CAPM context (adjusted for beta) or net of an array of accepted factors. As with Table 4, Panel B, we use a four-factor model, adding the momentum factor to the standard Fama-French (1993) beta, value, and size factors. Adjusted for the four-factor loadings, the alternatives that choose the top 500 by market-cap—with and without banding and seasoning—show little difference from the excess returns in Table 6, and they lack statistical significance. The alternatives that no longer anchor on current market-cap to choose additions and deletions, however, fare considerably better and show strong statistical significance.

We can also consider another perspective: It is not that the alternative selection methods add value but that the S&P 500 loses ground by not using a formulaic approach for selecting which 500 stocks to include in the index. The differences in performance

Table 6. Performance Attribution of Alternative Rules for Choosing 500 Stocks for Market-Cap-Weighted Indices, Using Fama-French Three-Factor Model Plus Momentum, October 1989–June 2021

		CAPM Model	Fama-French Three-Factor Model plus Momentum				
				Coefficients, Factor Loadings			
Index	Excess Return vs. S&P 500	Annual Alpha	Annual Alpha	Market[a]	Size	Value	Momentum
S&P 500	—	0.00%	0.00%	1.000	0.000	0.000	0.000
t-statistic	—	—	—	—	—	—	—
Top 500 by Market-Cap	0.25%	0.22%	0.20%	1.000	0.026	−0.049	0.008
t-statistic	*1.25*	*1.09*	*1.26*	*0.07*	*6.08***	*−10.96***	*2.56**
Top 500 by Five-Year Avg. Market-Cap	0.38%	0.43%	0.40%	0.988	0.014	−0.005	0.003
t-statistic	*2.88***	*3.24***	*3.05***	*−4.59***	*4.23***	*−1.44*	*1.30*
Top 500 by Fundamental Size	0.46%	0.58%	0.52%	0.980	−0.016	0.037	0.001
t-statistic	*2.63***	*3.46***	*3.54***	*−6.76***	*−4.14***	*9.04***	*0.52*
Top 500 by Market-Cap, with Banding & Seasoning at 20%	0.27%	0.26%	0.23%	0.998	0.018	−0.033	0.007
t-statistic	*1.66*	*1.56*	*1.60*	*−0.63*	*4.56***	*−8.16***	*2.56**
Top 500 by Five-Year Avg. Market-Cap, with Banding & Seasoning at 20%	0.41%	0.49%	0.44%	0.983	0.008	0.009	0.004
t-statistic	*3.03**	*3.83***	*3.42***	*−6.62***	*2.45**	*2.62***	*1.51*
Top 500 by Fundamental Size, with Banding & Seasoning at 20%	0.46%	0.58%	0.52%	0.980	−0.020	0.040	0.000
t-statistic	*2.54**	*3.43***	*3.52**	*−7.17***	*−4.28***	*9.67***	*0.74*

[a] For the market beta, the *t*-statistic measures significance of the difference from a neutral beta of 1.00.
** and * indicate significance at the 1% and 5% level, respectively. Please refer to text for descriptions of the strategies. We use the replicated S&P 500 return to proxy the market factor.
Source: Research Affiliates, LLC, based on data from Siblis Research, Wikipedia, and CRSP.

range from 25 bps to 46 bps a year; the CAPM beta-adjusted alpha ranges from 22 bps to 58 bps annually; and the four-factor alpha, after we adjust for the portfolios' style tilts relative to the S&P 500, ranges from 20 to 52 bps a year. Apart from the strategies that simply choose the 500 largest by market-cap—with or without banding and seasoning—the excess returns and alphas are highly statistically significant.

Figure 3 graphically shows the performance of these alternative cap-weighted indices, constructed using alternative selection criteria for the 500 stocks in the index fund, relative to the S&P 500. The simplest selection method is the 500 largest market-cap stocks, shown as the solid black line in all four panels. This simple largest-cap 500-stock portfolio has outpaced the S&P 500 by 25 bps a year since 1989 and by 20 bps a year after applying a four-factor model to capture the factor loadings of the portfolios. The dashed black line in all four panels shows the results if we apply banding and seasoning to our selection of the 500 largest-cap stocks.[42] Banding and seasoning not only reduce turnover by leaving out companies that jump onto the list and then quickly fall off the list but also add 2 to 3 bps of incremental return (or risk-adjusted alpha) relative to the simple top 500 strategy.

Panels A and C of Figure 3 show the performance of alternative 500-stock indices, constructed using formulaic methods that use no judgment in the selection process, relative to the S&P 500. Panels B and D show comparative results after adjusting for a four-factor model that captures the style tilts of these portfolios relative to the S&P 500. Depending on the selection method for the 500 stocks, the excess return gets a little better or a little worse, but becomes even more statistically significant in most cases.

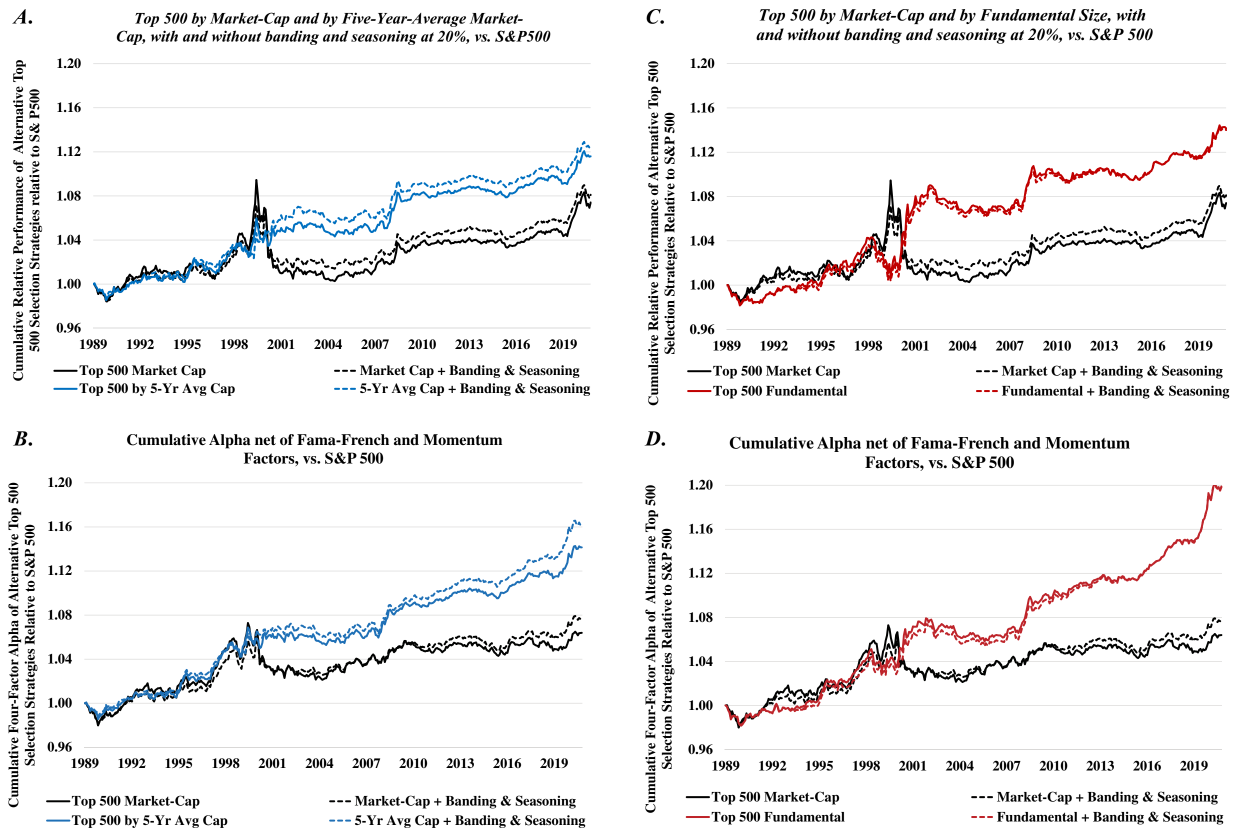
We compare the performance of an index constructed of the 500 stocks with the largest five-year-average market-cap, relative to the S&P 500, in blue in Panels A and B. The dashed line shows the results for both strategies with banding and seasoning at 20%. When we rely on five-year-average market-cap to select the 500 stocks (and better still add banding and seasoning to the selection process), we boost returns by another 13–14 bps, and the risk-adjusted alpha increases by another 20 bps or more.[43] When we use five-year-average market-cap, with banding and seasoning, our investor is 12.4% better off after 32 years than with the S&P 500. Adjusted for the four-factor risk model, the risk-adjusted excess return exceeds 16%. In the indexing world, these are big numbers. Note that we are not *weighting* by five-year-average market-cap. We are only *selecting* stocks on this basis, and then use simple cap-weighting to construct the portfolio.

In Panels C and D, the red lines show the performance of a cap-weighted portfolio of the 500 largest publicly traded businesses in the United States, selected using the four separate measures of company size we described earlier, relative to the S&P 500. The dashed lines show the application of banding and seasoning, which reduces portfolio turnover. Again, we are not weighting by fundamental size of the company, but by market-cap (the solid black lines reflect the same strategies shown in Panels A and B, the 500 largest market-cap stocks, with and without banding and seasoning). This simple expedient—choosing companies for the cap-weighted index fund based on the size of the business instead of the market value of the business—boosts performance by a noteworthy 46 bps a year (and by more than 50 bps a year when we adjust for the four-factor risk-profile differences relative to the S&P 500).

How important are these return differences? An investor in an index fund that simply identifies the 500 largest companies in the United States based on fundamental size of the business, then cap-weights them, is 14% wealthier than an investor in the S&P 500 (and 20% richer on a four-factor risk-adjusted basis). When we do not use market-cap to select the stocks for the index, banding and seasoning make essentially no difference to the return or risk profile of the portfolio, although these protocols do reduce turnover.

The benefits of using alternative selection methods for constructing the 500-stock index fund are, of course, episodic, but are reasonably reliable on a rolling five-year basis and are highly significant when adjusted for beta, size, value, and momentum. The biggest return differences—good and bad—occur mainly during the tech bubble of 1999 and in its aftermath. The indices that select stocks based on market-cap alone actually bought even *more* tech highfliers than the S&P Index Committee did, and bought them earlier. This boosted results relative to the S&P 500 during the tech bubble, which then reversed course as the bubble burst. The opposite happened for the indices constructed using the fundamental size of a company's business. These index fund strategies did not chase the tech bubble, so the S&P 500 sharply outperformed them until the bubble burst. When we examine our results net of a four-factor (beta, value, size, and momentum) risk model, the large swings before and after the tech bubble burst in 2000 largely disappear.

Figure 3. Cumulative Return of Alternative Top-500 Constituent Selection Method Relative to S&P 500, October 1989–June 2021.



Note: Please refer to the text for descriptions of the strategies.
Source: Research Affiliates, LLC, based on data from Compustat and CRSP.

Also, notable in the graphs, during the growth-dominated bull market of the 2010s, all of these strategies added less value relative to the S&P 500 than before or since. Panels B and D show us the reason. The strategies that select our 500 stocks based on the fundamental size of the business will exclude some highfliers if their underlying businesses are too small to make the 500-stock cut, and will include some larger companies that are deeply out of favor, hence, small-cap. So these strategies (with or without banding and seasoning) generally have a value bias relative to the S&P 500. Net of the factor loadings, the performance in the 2010s is little different from before and since. Risk-adjusted value added is reasonably relentless. The same dynamic drives the performance of the strategies that select the 500 largest stocks based on five-year average market-cap.

The strategies that select the 500 stocks based on current market-cap (with or without banding and seasoning) have a growth bias relative to the S&P 500, which on average excludes 120 of these largest-cap names. So, these strategies add less value relative to the S&P 500 once we adjust for the factor loadings from our four-factor risk model. Our work invites a question: Does the S&P Index Committee add value? From a pure return-and-risk perspective, maybe not, but we should not minimize the importance of *comfort*. The utility (in the academic finance meaning of the term) of having human involvement in an index's construction, even with no difference in performance, can have merit.

Nevertheless, our research points to ways the S&P Index Committee, and other index providers, can perhaps improve the performance of the *indexes*—concurrently lowering turnover and risk—that so many cap-weighted index fund managers are tracking. We see no reason why these alternative selection methods cannot become even more comfortable—higher utility—to the end customers, relative to the status quo. If, as we observe, the S&P 500 has a larger tracking error relative to the top 500 selected by market-cap than it does relative to the top 500 selected by fundamental size of the underlying companies' businesses, why, then, not do the latter?

Most studies of price impact document that transaction costs increase as the amount traded over a short time period increases, therefore implying that turnover-lowering techniques will reduce trading costs.[44] With five-year averaging of market capitalization, and applying both banding and seasoning, the resulting portfolio outperforms the S&P 500 by 41 bps a year from October 1989 to June 2021 and lowers annual turnover for the same period to 4.0%. When using company fundamental size as the stock-selection method, the strategy beats the S&P 500 by 46 bps from October 1989 to June 2021 with essentially the same turnover as the S&P 500. Combining fundamental-size selection with banding further reduces the turnover from 4.4% to 4.1% with the same 46 bps annual benefit relative to the S&P 500.

When index changes are delayed by either 3 months or 12 months as in the lazy replicated S&P 500, or when trading is accelerated to the announcement date or the day after, portfolio turnover obviously does not change. The other strategies we analyzed such as using a smoothed five-year-average market-cap and both 20% banding and seasoning to avoid pointless churning, are effective in lowering turnover and its associated costs.

Table 6 vividly illustrates the efficacy of this approach. We find that with each additional step of (1) averaging historical market-cap to select stocks, (2) banding and seasoning to inhibit pointless trades, and (3) applying both approaches in one strategy, investors can add value while reducing risk. Selecting index constituents by fundamental company size, with the application of banding and seasoning, is even more powerful, with the highest value-add and one of the lowest levels of turnover of any of the strategies we tested.

Putting the Pieces Together

Investing in traditional cap-weighted indices has a negative impact on portfolio returns that arises from three sources: mean reversion, turnover, and selection bias. *Mean reversion* is a result of additions to the index that tend to be newly expensive and typically move downward in price after they are added, while discretionary deletions tend to be newly cheap and typically rebound in price after their deletion. *Turnover*, and all of its associated trading costs, is neatly hidden by delaying the index change until after the index funds have had an opportunity to trade them. *Selection bias*, which is the tacit choice to add or drop stocks based on current valuation multiples and recent momentum, can increase an index fund investor's exposure to growth and momentum.

All three problems garner little attention because of index providers' preannouncing trades, offering index fund managers a grace period to complete their trades before the index is officially changed, so the trading appears to be free. All three problems can be mitigated if we seek to neutralize the selection problem by patiently reconstituting the index based on valuation-indifferent measures of company size or smoothed measures of market-cap and then apply banding and seasoning to avoid the flip-flops.

We have shown that lazy trading, up to a year after a change has been announced, adds material and statistically significant alpha. In an efficient market, this should not happen. In an inefficient market, if the trading of index funds moves share prices away from their (unknowable) fair value and the market's quest for fair value then leads to mean reversion, then lazy trading should improve performance.

We have shown that anticipating index changes is a worthwhile enterprise. Adding the performance spreads between additions and deletions, from the day before the trade is announced to the day after the trade date (when most index funds make their trades), results in a performance spread between additions and discretionary deletions of more than 1,600 bps. Correct anticipation of index trades is clearly amply worthwhile.

We have shown that delayed index construction, based on a five-year-average market-cap or on company fundamental size, with banding and seasoning applied in both strategies to minimize the risk of flip-flops (additions that are quickly deleted), can materially improve performance, although tracking error relative to current indices may discourage some investors. *If the change in construction method was also made by the index providers, however, performance could be improved without index fund managers incurring any additional tracking error because the index itself changed!*

Conclusions

Sharpe (1991, 7) asserted that the logic of traditional cap-weighted index fund investing relies on the assumption that "before costs, the return on the average actively managed dollar will equal the return on the average passively managed dollar." This assumption, while approximately correct, relies on another assumption, that passive investors hold the same collection of securities as the totality of active managers *and do not trade*. Unfortunately, in practice, this assumption does not hold for investors in typical large-cap index funds.

George Box famously observed that "all models are wrong, but some models are useful." Neither Sharpe's CAPM nor his "Arithmetic of Active Management" (1991) is precisely correct, but both are assuredly useful. They become more useful if we are willing to acknowledge their limitations and seek ways to recapture avoidable costs in the indexing process.

Quoting Berk and van Binsbergen (2015, 10):

> What Sharpe's argument ignores is that even a passive investor must trade at least twice, once to get into the passive position and once to get out of the position. If we assume that active investors are better informed than passive, then whenever these liquidity trades are made with an active investor, in expectation, the passive investor must lose and the active must gain. Hence, the expected return to active investors must exceed the return to passive investors, that is, active investors earn a liquidity premium.

Berk and van Binsbergen, and others,[45] are arguing that active managers, on average, should outperform passive managers (net of style exposures, but before fees and implementation costs!) by providing them with liquidity, taking the other side of the indexers' trade by buying the potentially undervalued stocks that index funds sell and selling the potentially overvalued stocks that index funds buy. Unfortunately, active management fees and trading costs may be larger, perhaps much larger, than the aggregate "liquidity rent" collected by the active managers. This does not mean that all active managers should be expected to lose, only that the winning active managers need a loser on the other side of their trades.

Our research supports this view by suggesting expedients which can help an index fund manager earn modest above-market returns by trading early, immediately after the change in the index is announced, or late, trading 3 to 12 months after the index is changed, or by choosing stocks for a cap-weighted index fund based on metrics that mitigate the tendency to buy highfliers and sell feared and loathed deep-value stocks that are newly cheap. The loser on the other side of these trades is the conventional indexer, willing to sacrifice performance in order to lock in zero tracking error.

We argue that index providers (S&P, MSCI, FTSE, and so on) can construct better-performing indices that are less prone to chasing recent performance and that have lower turnover. In so doing, index providers would stop offering a free lunch to active managers. The use of five-year-average (or longer!) market capitalization to identify the more stable and reliable top 500 or top 1,000 companies, as well as the use of banding and seasoning to inhibit flip-flop trades (stocks added to the index that are quickly deleted), would largely eliminate the buy-high/sell-low dynamics of standard large-cap indices, reduce portfolio risk, and improve portfolio returns, all while reducing index fund turnover. If we select companies based on the size of their business, then weight them by market-cap, we fare better still.

The return drag of the traditional indices, and the opportunity to avoid it, is largely a consequence of the obsession with tight tracking error when measuring index fund performance. We cannot add value without tracking error. Investors should be more holistic in setting and monitoring their risk-and-return objectives. It makes no sense to tolerate (even seek) large tracking error for parts of our portfolio and no tracking error for others. Likewise, index providers have a wonderful opportunity to revisit their rebalancing rules in order to recapture some of the rebalancing costs they inadvertently impose on their customers.

Editor's Note

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Notes

1. This finding is not new. The attribution of return differences accumulated over the year before and after an index change *is* new, as is the use of selection metrics, other than current market-cap, for index construction.
2. The transaction costs passed on to investors are far from trivial. For example, Gastineau (29 2002) estimated S&P 500 Index transaction costs to be 50 to 100 bps and for the Russell 2000 to be 200 to 300 bps. The problem can be accentuated for the ETF managers tasked with closely

tracking the underlying indices. For example, Li (29 2021) found that ETF managers place much of the trading positions at the closing price of the reconstitution day. This leads to an estimated 67 bps of execution costs, which he estimates to be about three times larger than the average cost incurred by institutional investors.

3. For the purposes of this article, we distinguish between deliberate additions and deletions that arise due to explicit decisions of dthe governing committee or index rule (we label these *discretionary*) and those occurring due to corporate actions such as merger, spin-offs, or acquisitions (we label these *nondiscretionary*).

4. Pavlova and Sikorskaya (2022) documented a similar outcome for stocks around the Russell 1000/2000 cutoff. They show that stocks with increased index holdings tend to be negatively correlated with the future return.

5. An argument could be made that if all index managers were to embrace our suggestions, our own variants of the strategy would have large trading costs and thus the costs we identify as avoidable might no longer be avoidable. However, if not all indexers were to trade the same lists at the same times, then the remarkably large trading costs we demonstrate can be avoided, at least in part.

6. Managers do deviate from the index they replicate to respond to the incentives provided by the market. For example, Gastineau (2004) found that ETF managers disproportionally favor trades to improve their fund's after-tax performance in response to investors' choosing an ETF as a tax-efficient index-implementation vehicle.

7. Historical annual turnover numbers from 1993 to 2021 are based on data from Factset. Annual turnover estimates from 1989 to 1992 are based on our own replication of the S&P 500.

8. The annualized dollar amount of trading is $2 \times 4.4\% \times \$7.1$ trillion $= \$625$ billion. Whereas most of the trading is concentrated in the additions and deletions, some trading—especially on the sell side of the trade—is spread across the remaining almost 500 stocks. This occurs because the stock positions added are almost always far larger than the stock positions sold. For example, when Tesla (TSLA) replaced Apartment Investment & Management Co. (AIV), indexers had to invest 1.69% of their assets in TSLA, while the sale of AIV amounted to a scant 0.01% of the S&P 500. The remaining 1.68% of sales would have been prorated across all other index constituents.

9. According to the S&P Dow Jones Indices *Annual Survey of Assets*, as of year-end 2021 the benchmarked assets were around $8.5 trillion.

10. For example, Chinco and Sammon (2021) documented a significant spike in volume around index rebalancing dates.

11. We are not suggesting that this is in any way nefarious. Index funds are the fee-paying customers of index calculators. The index calculators are simply serving their clients. That said, we do object to suggestions that trading costs are zero.

12. This is not a hypothetical assertion. Some efforts to add value with early trading have been shut down, after years of success, at the behest of client-facing and marketing teams, tired of fielding questions about tracking errors from clients and client prospects, even when the tracking error has been positive.

13. We cannot know, of course, how index fund managers would have altered their behavior or how their behavior might have affected the trajectory of subsequent returns should the S&P Index Committee not have changed its policy in 1989. Thus, we describe these results as hypothetical. If it were not for the obsession with zero tracking error, a 20 bps shortfall, with 31 bps tracking error, gives an enormous negative information ratio of −0.64. It is difficult to imagine that index funds would not now be vying to better one another on mitigating these trading costs rather than trading *en masse* in an immense market-on-close block trade at the end of the trade date.

14. We do not have access to the official data from Standard & Poor's, so we reconstructed the index history from publicly available sources. The sample was constructed based on data from Siblis Research and Wikipedia. The raw constituent change list from Siblis Research includes company names, tickers, action of the change (addition or deletion), announcement date, and effective date. We used constituent change data from Siblis for the period March 1973 to March 2017 and data from Wikipedia and S&P press releases for the period April 2017 to June 2021.

15. The number of constituents in the S&P 500 is not always 500. Since 2015, the index has followed a methodology change that allowed multiple share classes of S&P 500 constituent companies. At the time of final edits to this article, the S&P 500 comprises 503 stocks.

16. We categorize as nondiscretionary the additions that lack price data in the six months before the trade date and the deletions that lack price data for the six months after the trade date. This rule will not mistakenly characterize recent IPOs as nondiscretionary additions because, according to *S&P U.S. Indices Methodology* (2022), an IPO is required to have at least 12 months of history to be eligible for consideration for S&P 500 inclusion.

17. We intend to examine bankruptcies, as a special case, in a future study. These stocks are deleted from the index after the share price fully reflects post-bankruptcy bleak expectations and are further depressed by index fund sales. These stocks may very well be priced with an outsized risk premium and therefore subsequently perform very well.

18. The effect may be surprising in magnitude and statistical significance, but is not uncharacteristic of the largely nontradeable signals such as index inclusion.

19. The preannouncement, or grace period, return is partly driven by index fund managers who are willing to accept some tracking error in order to begin early purchases of the additions and sales of the deletions, and partly by liquidity providers who accumulate shares of additions (and short positions in the pending deletions) in order to provide liquidity to the index funds on the trade date. Of course, other factors, such as improved analyst coverage

or an increase in expected future liquidity, may contribute to the price moves during the grace period. Another important feature that potentially contributes to the return dynamics is the lower risk profile of the larger and often more familiar companies added to the index and the higher risk profile of the deletions. Investors may be more comfortable with these index choices and thus are willing to bear the lower consequent returns. This observation is consistent with the findings of Elliott et al. (2006), who attribute much of the addition effect to increased market awareness. Our statistical tests suggest the mean-reversion effect is stronger than either the liquidity or risk profile effects, but does not subsume the latter.

20. Cynics might suspect that word leaks out about the pending change in the index. We believe a more plausible explanation is that hedge funds (and some index fund managers) make educated (and correct) guesses as to likely index changes.

21. We observe an asymmetry in the returns after the trade date, for which deletions are responsible for a large portion of the difference. We would speculate that the deleted stocks are relatively illiquid and sport an outsized risk premium after deletion, leading to a much larger mean-reversion return after their removal from the index.

22. This asymmetry of return between additions and deletions is consistent with the findings of Chan, Kot, and Tang (2013) who observed a permanent positive effect on the stock price from being included in the S&P 500. They attributed the effect to the likelihood of broader coverage by analysts, higher liquidity, and more institutional ownership.

23. Our article also contributes to a long-running debate in the academic literature about equity demand elasticity. Specifically, starting with Shleifer (1986), researchers have used index rebalancing to study equity demand slope. We showed that the demand slope has remained downward sloping, albeit much of the impact has shifted to the period between the announcement and the actual index rebalancing. This is consistent with Pavlova and Sikorskaya (2022), who reached a similar conclusion for Russell indices rebalancing. It is also consistent with Haddad, Huebner, and Loualiche (2022), whose main conclusion is that the demand has become more inelastic with the proliferation of indexing in the last decades. Madhavan, Ribando, and Udevbulu (2022) analyze the recent Russell index rebalancing and find high liquidity provision in the recent periods, in which the costs of liquidity provision, as measured by the beta-adjusted temporary impact, can actually be negative.

24. The valuation discount is defined as the valuation ratio of the stock relative to the market valuation ratio. With the exception of P/B, for which we use single-year book value, we use five-year-average fundamental values in the other price-to-fundamentals ratios: P/E, P/CF, P/S, and P/D. The average valuation discount is the exponent of the pooled average of the log of the valuation discount ratio for each individual metric. The total average is the simple average of the four.

25. These are stand-alone calculations we do not report separately in the tables.

26. $0.16\% + 4.99\% - 1.61\% = 3.54\%$

27. $-2.20\% - 7.23\% + 20.43\% = 11.00\%$

28. For replication, we used data on the constituent changes (i.e., the additions and deletions to the index) from Siblis Research (October 1989–March 2017) and Wikipedia (April 2017–June 2021), paired with price and market capitalization data from CRSP.

29. For periodic cross-checking, we used SPY ETF holdings from December 2010 and September 2005 (when the S&P 500 fully transitioned to float-adjusted weighting). We used Vanguard 500 holdings as of December 1999. The holdings data are from Bloomberg.

30. The replication has a cumulative annualized average return within 0.04% of the actual index.

31. Arnott, Beck, and Kalesnik (2015) found a portfolio that uses 20-year-old stale S&P 500 holdings and weights, leaving out the companies that no longer exist, outperforms the S&P 500 by about 180 bps a year.

32. Lazy strategies inherently have a small bias toward value and small-cap, because they trade after some mean reversion has occurred. The latter strategy was first identified by Banz (1981), who showed that small-cap stocks outperform large-cap stocks. This phenomenon is known as the size premium. Berk (1997) argued theoretically and demonstrated empirically that the small-cap effect is largely driven by small-cap stocks being relatively cheaper and large-cap stocks being relatively more expensive.

33. Of course, for an investor to capture all of the 20 bps may prove difficult, because it would require significant trading and the costs of trading would detract from the performance.

34. In “Reports of Value’s Death May Be Greatly Exaggerated” by Arnott et al. (2021), the authors showed that value’s underperformance in the period since 2007 can be entirely explained by value’s getting increasingly cheap, using a blend of relative valuation multiples, relative to growth. At the end of their sample, in the summer of 2020, value was at its cheapest valuation in history. Because valuations have historically been mean-reverting, that strongly implied a high potential for the value style to outperform going forward. Value outperformance should translate into the tailwind for the delayed strategies that we examine here as well.

35. Arnott, Beck, and Kalesnik (2015) used stale index weights from 5, 10, and even 20 years earlier to create an index fund that earned up to 180 bps in excess return with lower-than-market risk and strong statistical significance. Of course, the tracking error of this type of strategy would be large, partly because the holdings are very different from the S&P 500 and, more significantly, because the weights are very different.

36. When less than five years of history are available, we use the time span that is available. In this way, an IPO that immediately ranks in the top 500 will not need to wait five years to be included.

37. This is essentially a more modern version of the Fundamental Index, originally described by Arnott, Hsu, and Moore (2005), which is then reweighted by market-cap.

38. This approach sometimes leads to having more stocks to add than to drop, and vice versa. We correct for this by allowing stocks with ranks from 400 to 600 to enter or exit the portfolio, as necessary, in order to maintain the target of 500 stocks in the portfolio. If, for example, 20 stocks have risen above rank 400 and are not in the index, while 30 stocks in the index have fallen below rank 600, we can choose to drop just the 20 smallest stocks so that the index remains a 500-stock index.

39. Indeed, just 6.8% of the fundamentally selected cap-weighted index is invested in stocks that are not in the S&P 500. Reciprocally, 2.3% of the S&P 500 is invested in stocks that are not in the 500 largest U.S. companies, selected by fundamental size of the business. Of this 2.3% difference, nearly half is invested in stocks of companies that are not domiciled in the United States and therefore are ineligible based on our criteria.

40. If we excluded IPOs and other stocks with less than three years of market-capitalization observations, the index performance would improve by another 5 bps in the simulations with no turnover control and by 2 bps with turnover controls. Excluding recent IPOs would also increase tracking error from 75 bps to 126 bps, making it, practically, less attractive.

41. None of these figures reflects trading costs. If turnover is lowered by banding and seasoning, it is possible that the performance net of trading costs would improve in all three cases.

42. Many academic papers on investment factors and anomalies show tables of data that span years or decades but do not include graphical illustrations that show the evolution of the efficacy of their strategies. We believe this is a mistake because graphs of value-add or alpha can instantly highlight strategies that are particularly episodic or are losing efficacy. As one noteworthy example, standard momentum has added zero excess returns from 1999 through year-end 2021 (see, for example, Arnott et al. 2017).

43. These tens of basis points matter. The S&P 500 investor is 6.9% to 11.1% poorer, after just under 32 years, than an investor in index funds that are tied to these alternative mechanistic stock-selection methods. To be sure, this difference ignores trading costs and other forms of implementation shortfall. But, on modest assets, there is no reason to believe that these costs would be large, given the very low turnover of all of these strategies.

44. The same finding also implies that the commonly used approach of rebalancing annually on a single day of the year is flawed.

45. Pedersen (2018) examines in details the arithmetic of active and passive investor interaction.

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